

MPG No. 8.9.128 $m'm'm$ (mm'm' setting) [Type III, orthorhombic] [M tensor]

* Rank 0 tensor. * Rank 1 tensor.

$$\begin{bmatrix} M_x & 0 & 0 \end{bmatrix}$$

$$M_x = M_{px}^{(1)}$$

* Rank 2 tensor (s). * Rank 2 tensor (a). * Rank 3 tensor (ss).

$$\begin{bmatrix} M_{xxx} & 0 & 0 \\ M_{yyx} & 0 & 0 \\ M_{zzx} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & M_{zzx} \\ 0 & M_{yyx} & 0 \end{bmatrix}$$

$$M_{xxx} = 2M_{fax}^{(1)} + 3M_{px}^{(1)}$$

$$M_{yyx} = -M_{fax}^{(1)} + M_{fbx}^{(1)} + M_{px}^{(1)}$$

$$M_{zzx} = -M_{fax}^{(1)} - M_{fbx}^{(1)} + M_{px}^{(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ M_{yyx} & 0 & 0 \\ M_{zzx} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -\frac{M_{zzx}}{2} \\ 0 & -\frac{M_{yyx}}{2} & 0 \end{bmatrix}$$

$$M_{yyx} = 2M_{px}^{(2)} + 2T_{dyz}^{(1)}$$

$$M_{zzx} = 2M_{px}^{(2)} - 2T_{dyz}^{(1)}$$

$$\frac{M_{zzx}}{2} = M_{px}^{(2)} - T_{dyz}^{(1)}$$

$$\frac{M_{yyx}}{2} = M_{px}^{(2)} + T_{dyz}^{(1)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & M_{zzz} \\ 0 & M_{xyy} & 0 \end{bmatrix}$$

$$M_{zzz} = M_{px}^{(3)} + T_{dyz}^{(2)}$$

$$M_{xyy} = -M_{px}^{(3)} + T_{dyz}^{(2)}$$

* Rank 4 tensor (sss). * Rank 4 tensor (ssa). * Rank 4 tensor (aas). * Rank 4 tensor (aaa). * Rank 4 tensor (sa). * Rank 4 tensor (as). * Rank 4 tensor (s). * Rank 4 tensor (a). * Rank 4 tensor (t).